

APM 503: Applied Analysis

Arizona State University
School of Mathematical and Statistical Sciences
Fall 2026

Course Information

Instructor	Prof. Jimmie Adriaola
Email	jimmie.adriaola@asu.edu
Office	GWC 651
Office Hours	TBD and by appointment
Meeting Time	TBD
Classroom	TBD
Prerequisites	Advanced calculus, linear algebra, and differential equations, or instructor permission

Course Description

This course is a first graduate course in applied analysis designed to prepare students for further study in partial differential equations, harmonic analysis, integral equations, and the calculus of variations. The course begins with a concise but rigorous review of the real number system and classical analysis from a modern point of view, then develops measure and integration, L^p spaces and duality, Hilbert space methods, approximation by smooth functions, Schwartz space and tempered distributions, and a first serious introduction to Fourier analysis and harmonic analysis. The emphasis throughout is on conceptual clarity, proof, and the analytic structures that recur across modern applied mathematics.

A central goal of the course is to give students a solid working command of the spaces, convergence principles, approximation methods, and transform techniques that underlie advanced analysis. The course is intended to end at the threshold of more specialized topics rather than attempt to cover them prematurely. In particular, the material is meant to set students up well for later courses in PDEs, weak and variational methods, integral equations, and advanced applied analysis.

Texts and References

There is no single required text. We will draw from the following references throughout the semester.

- John K. Hunter and Bruno Nachtergaele, *Applied Analysis*
- Elliott H. Lieb and Michael Loss, *Analysis*
- Elias M. Stein and Rami Shakarchi, *Fourier Analysis: An Introduction* and *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*
- Tom M. Apostol, *Mathematical Analysis: A Modern Approach to Advanced Calculus*

Topics to Be Covered

The precise pacing may vary, but the course is expected to cover most of the following topics.

1. The real numbers, completeness, supremum and infimum, sequences, subsequences, series, and compactness in $\mathbb{R}^n$
2. Continuity, differentiation, and integration from a rigorous modern point of view
3. Metric spaces, normed spaces, and examples arising in analysis
4. Sigma-algebras, measurable functions, Lebesgue measure, and Lebesgue integration
5. Convergence theorems, including monotone convergence, Fatou's lemma, dominated convergence, and Fubini–Tonelli
6. L^p spaces, Hölder's inequality, Minkowski's inequality, completeness, density, and duality
7. Hilbert spaces, orthogonality, orthonormal systems, projection, and the Riesz representation theorem
8. Convolution, approximate identities, mollifiers, and approximation by smooth functions
9. Schwartz space $\mathcal{S}(\mathbb{R}^n)$ and tempered distributions $\mathcal{S}'(\mathbb{R}^n)$
10. Distributional derivatives and the passage from classical to weak analytic formulations
11. Fourier transform on rapidly decaying functions, L^2 methods, and first extensions to tempered distributions
12. Selected topics in harmonic analysis, including convolution operators, inversion principles, Plancherel theory, and basic transform methods

Tentative Semester Arc

Weeks	Primary Themes
1–2	Module 0: real numbers, completeness, sequences, compactness, continuity, differentiation, integration, and classical proof techniques
3–4	Measure spaces, measurable functions, Lebesgue measure, and Lebesgue integration
5–6	Convergence theorems and L^p spaces, including Hölder, Minkowski, completeness, density, and duality
7–8	Hilbert spaces, orthogonality, projection, orthonormal bases, and the Riesz representation theorem
9	Convolution, smooth approximation, approximate identities, and mollifiers
10–11	Schwartz space, tempered distributions, and distributional derivatives
12–14	Fourier transform, L^2 methods, and harmonic analysis as the capstone of the course
15	Synthesis and further directions: preparation for PDEs, integral equations, and variational analysis

Course Goals

By the end of the course, students should be able to work carefully with the foundational spaces of analysis, use measure-theoretic convergence tools correctly, reason fluently in L^p and Hilbert space settings, understand approximation by smooth functions and the language of distributions, and use Fourier-analytic ideas as a unifying framework for further study in advanced applied analysis.

General Remarks

This is a proof-based course. Students are expected to engage seriously with definitions, examples, counterexamples, and theorems, and to move comfortably between concrete calculations and abstract formulations. The syllabus is a working document and may be adjusted as the semester develops.